



APPLICATION FOR MASTER'S-LEVEL AFFILIATED FELLOWSHIPS 2026-2027 COMPETITION

(Application Deadline: 1 December 2025 before 5:00 pm PST)

Applicant Information

Last name of applicant Klukas	First name of applicant Dominic	Initial(s) of middle name(s) H.
UBC student number 64348378	Email klukasdominic@gmail.com	

Degree Information

UBC department / graduate program as of September 2025 (e.g., Chemistry, Asian Studies) Electrical and Computer Engineering	Degree type in which I am registered as of September 2025 (e.g., MSc, MASc, PhD) MASc
Proposed UBC department / graduate program 2026-2027 academic year (e.g., Chemistry, Asian Studies) Electrical and Computer Engineering	Proposed degree type in 2026-2027 academic year (e.g., MSc, MASc, PhD) MASc
Research Subject Area (see Instructions for definitions) Research Subject Area 1, Electrical Engineering	
As of December 31, 2025, I will have completed 3 months of Master's study (total at UBC and other institutions)	
As of December 31, 2025, I will have completed 0 months of doctoral study (total at UBC and other institutions)	

Citizenship

Citizenship Canada
I have applied for 2026-2027 funding from one of the following three federal funding agencies: ☐ CIHR ☒ NSERC ☐ SSHRC ☐ I have not applied for 2026-2027 funding from one of the three federal funding agencies because (provide explanation in the field below):

Signature

I hereby agree that to the terms and conditions outlined in the "Signature" section of the application instructions. Dominic Klukas _____ Applicant's signature Please type your full name above and we will accept that as your signature.	2025-11-27 _____ Date
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Summary of Proposal

Lay Title (99 characters maximum)

Equivariant Reinforcement Learning Policies with Symmetry Breaking

Summary (1,800 characters maximum)

Deep Reinforcement Learning (RL) is an AI algorithm that offers promising solutions to decision and control problems such as robotics tasks. While promising, it typically requires a huge amount of training data and often struggles when the situation changes even a little bit from what it saw during training.

Part of the problem is that standard Deep RL methods don't make use of many of the simple patterns that exist in the real world. For example, many robotics tasks, such as picking up objects and moving them, stay basically the same if the original location of the object is translated or rotated relative to the robot. These patterns are called symmetries, and using them can make learning much faster and more reliable.

When so-called equivariant neural networks are used in Deep RL, the symmetries are built into the algorithm, and automatically respected. However, often symmetries are partially broken in the real world. For example, when picking up an object in one location, we may have to avoid an obstacle not present anywhere else. Because equivariant networks enforce symmetry everywhere, they can't adapt to these exceptions.

In my project I will address this key challenge with three objectives. First, I aim to develop algorithms using mathematical analysis that can detect regions in a state space where symmetries are broken in a robot trajectory dataset. Next, I will design new equivariant neural networks that relax symmetries in specific regions, combining efficient learning when symmetry holds with specialized behavior when needed. Finally, I will classify real robotics tasks and empirically determine when equivariant algorithms, equivariant algorithms with symmetry breaking, and standard Deep RL algorithms perform best.

Sex- and Gender-Based Analysis

Are sex (biological) considerations taken into account in this proposal?

☐ Yes

☒ No

Are gender (socio-cultural) considerations taken into account in this proposal?

☐ Yes

☒ No

If you selected 'yes' to either of the above questions, describe how sex and/or gender considerations will be considered in your program of study. (1,400 characters maximum)

Keywords

List up to 10 keywords that best describe the proposal (40 characters maximum per entry).

1. AI
2. Robotics
3. Reinforcement Learning
4. Geometric Deep Learning
5. Equivariance
6. Symmetries
7.
8.
9.
10.

CRITERIA-BASED AFFILIATED FELLOWSHIPS CHECKLIST

Review the descriptions of the criteria-based awards and indicate those awards for which you are eligible from the list by using the checkboxes.

Note that, in addition to the 'criteria-based' awards noted below, there are numerous 'open/unrestricted' awards that applicants will automatically be considered for.

The award values listed below are subject to change for the 2026/27 academic year.

☐ **#145 - T.K. Lee Scholarship**

A \$1,500 scholarship endowed by T. (Ting) K. Lee is awarded on the recommendation of the Faculty of Graduate and Postdoctoral Studies to a student from the People's Republic of China. Financial circumstances will be considered.

☐ **#146 - Basant Singh and Harnam Kaur Singh Fellowship**

A \$9,400 fellowship is awarded to a full-time student in a Master's or Doctoral program in any discipline, whose work will contribute directly or indirectly to a better understanding and the ultimate advancement of immigrant visible minorities in Canada.

☐ **#334 - Chiu Lin and Grace Ho Memorial Graduate Scholarship**

Scholarships totalling \$27,479 have been made available through an endowment established by an estate gift from Chiu Lin Ho for outstanding international graduate students entering their first year of study in any Faculty. Preference will be given to international students from China, Taiwan or Hong Kong. *Please note: Nominees for this award should be "entering their first year" in September 2025.*

☐ **#350 - Tina and Morris Wagner Foundation Fellowship**

This fund, established through a bequest from Mr. and Mrs. Morris Wagner, provides annual income of \$79,250. The income may be used to provide loans or fellowships for students in the Humanities.

☒ **#484 - Dr. Chi-Kit Wat Scholarship**

Two scholarships of \$3,050 each have been endowed by family and friends in memory of Dr. Chi-Kit Wat. Dr. Wat was associated with the University of British Columbia for many years, both as a student and, later in her career, as a research scientist. One award is made to a visa student from the People's Republic of China who is pursuing post-baccalaureate studies and demonstrates outstanding academic achievement and promise. A second award is made to a Canadian citizen or Permanent Resident of similar academic calibre.

☐ **#486 - C. L. Wang Memorial Scholarship**

A \$1,150 scholarship has been endowed in memory of C.L. Wang by family and friends. C. L. Wang, born Wang Chang-Lin in 1902, was the son of a mandarin official in the last Chinese dynasty before the revolution in 1911. He graduated from Tsing Hua University in Beijing in 1921. With a Tsing Hua scholarship, he studied at the University of Chicago completing his B.Com. in Economics. Following graduation, he worked at the Harris Trust Co. in Chicago, and returned to China in 1926. His career in banking spanned nearly four decades. He was the manager and a Director of the Shanghai Commercial Bank until he retired in 1964. He firmly believed in education and encouraged the young not only in the field of economics and banking to which he devoted his life, but also in engineering and medicine. He died a peaceful death in 1987, at the age of 85. The award is made on the recommendation of the Faculty of Graduate and Postdoctoral Studies to a graduate student from the People's Republic of China, with preference given to a student in Commerce and Business Administration, Engineering or Medicine.

☐ **#527 - Hugo E. Meilicke Memorial Fellowship**

This fund was established by the late Hugo E. Meilicke who gave distinguished service to the community through his business associations and through active participation in organizations such as the Kiwanis Club, the Salvation Army, the Vancouver Foundation, the Crippled Children's Hospital, the Vancouver Art Gallery, and the Vancouver Symphony Society. The annual income of \$44,450 provides awards in various fields such as agriculture, political science, commerce, fine arts, and music.

#1617 - Bank of Montreal Graduate Fellowship

Fellowships totaling \$32,900, endowed by the Bank of Montreal, is available to outstanding graduates in any field but preference is given to graduate students in the field of finance and economics.

☐ **Applying for #1617-finance**

☐ **Applying for #1617-economics**

(students in other fields will be considered automatically)

☐ **#4436 - Werner and Hildegard Hesse Fellowship in Ornithology**

Fellowships totalling \$50,200 have been endowed through a bequest by Werner Hans Hermann Hesse for graduate students engaged in ornithological research projects, with preference given to those studying wild birds. Werner and Hildegard Hesses' passion for wild birds was sparked in night classes on the ecology and conservation of birds at UBC and led them to become leaders in amateur ornithology. The Hesses conducted bird surveys in the Canadian arctic, compiled BC's Christmas Bird Counts for over 20 years, and developed a special concern for the harmful effects of human development on bird habitats and populations. The Hesses delighted in funding ornithological research and, particularly, contributing to the long-term monitoring of population change in seabirds of the Pacific Coast.

☐ **#4811 - Westcoast Energy Inc. Jack Davis Scholarship in Energy Studies**

A \$5,300 scholarship has been endowed by Westcoast Energy Inc. in memory of the Honourable Jack Davis. The award is made to a graduate student in energy studies.

☐ **#4820 - Du Pont Canada Fellowship in Pulp and Paper**

A \$6,300 fellowship has been endowed by Du Pont Canada Inc. and the Province of British Columbia. The award is offered to a graduate student undertaking research related to the pulp and paper industry.

☐ **#4833 - Shaughnessy Hospital Volunteer Society Fellowship in Health Care**

Two \$10,575 fellowships have been endowed by the Shaughnessy Hospital Volunteer Society for graduate students in the field of health care. Preference may be given to students with volunteer experience.

☐ **#4856 - Gertrude Langridge Graduate Scholarship in Humanities**

A scholarship of \$5,600 has been endowed through the bequest of Gertrude Langridge. The award is offered to a graduate student in any field of Humanities.

☐ **#4857 - Gertrude Langridge Graduate Scholarship in Medical Sciences**

A scholarship of \$5,900 has been endowed through the bequest of Gertrude Langridge. The award is offered to a graduate student in any field of Medical Sciences.

☐ **#6326 - C.D. HOWE Graduate Fellowship in Public Policy**

Fellowships totalling \$21,300 fellowship has been endowed by The C.D. Howe Memorial Foundation and The University of British Columbia. The award is offered to a graduate student in public policy.

☐ **#6336 - Anne and John BROWN Fellowship in Diabetes and Obesity Related Research**

Fellowships totalling \$24,000 have been endowed by Ann and John Brown. The awards are offered to graduate students involved in diabetes and obesity related research.

#6350 - Cordula and Gunter Paetzold Fellowship



Fellowships of \$18,000 each have been endowed by Cordula Paetzold and the University of British Columbia for full-time masters or doctoral students, with preference given to graduate students with disabilities, First Nations graduate students, and women students pursuing graduate study in medicine, neuroscience, genetics, health care and epidemiology, pharmaceutical sciences, music, theatre and film, forestry and architecture. If, in any given year, there are not sufficient eligible candidates among the preferred groups, awards may be granted to outstanding graduate students in any field. The fellowships may be renewed for a second year, subject to satisfactory academic standing, attendance and progress.

☐ **#6350 - disability**

☐ **#6350 - First Nations**

☐ **#6350 - woman**

(other eligible students will be considered automatically)

☐ **#6364 - Gerhard Henrik Armauer-Hansen Memorial Scholarship**

Scholarships totalling \$6,450 have been endowed in memory of Dr. Gerhard Henrik Armauer-Hansen, through a bequest by Mary Armauer-Hansen, for graduate students in the Faculty of Medicine or the Department of Microbiology and Immunology who are pursuing research into Hansen's Disease (leprosy) or other bacterial diseases.

☐ **#6391 – Michael W. Stahl Memorial Graduate Scholarship**

A \$1,450 scholarship is offered by Guenter and Diane Stahl in memory of their son, Michael W. Stahl (March 16, 1969 to November 9, 2005), to a graduate student conducting research in the areas of addiction, depression, or mental illness. In spite of some progress from recently initiated treatment for depression, Michael died after many years of struggling with alcohol and drug addiction. The award is made on the recommendation of the Faculty of Graduate and Postdoctoral Studies.

☐ **#6394 - Frieda Granot Graduate Scholarship in Interdisciplinary Research**

A \$400 scholarship has been endowed in honour of Dr. Frieda Granot to recognize over thirty years of service to The University of British Columbia, both as a professor in the Sauder School of Business, starting in 1975, and as Dean of the Faculty of Graduate Studies (1996-2006). The award is made to a Ph.D. student conducting interdisciplinary research.

☐ **#6451 - Pei-Huang Tung and Tan-Wen Tung Graduate Fellowship**

Fellowships totalling \$68,100 have been endowed by Mr. Pei-Huang Tung for international graduate students who are citizens of the People's Republic of China or Taiwan. Fellowships may be renewed for a second year of study at UBC subject to recipients maintaining good academic standing. In adjudicating eligibility, the financial circumstances of eligible candidates may be taken into account.

☐ **#6495 - Andrew Nord Fellowships in Rheumatology**

Fellowships totalling \$73,333 have been endowed through a bequest from the Estate of Andrew Nord. Andrew Nord suffered from the effects of arthritis from the age of 15 and the goal for his bequest is "to prevent the disease from its devastating onslaught, to find better ways to live with the disease, and to find the cause, leading to a cure once and for all." Doctoral students engaged in rheumatology research in any faculty at UBC are eligible to apply for the fellowships.

☐ **#6498 - Elwyn Gregg Memorial Fellowship**

Fellowships totalling \$34,300 have been endowed by a bequest from the Estate of Elwyn Gregg for graduate students in any field of study. Mr. Gregg believed that too often the best graduate talent from UBC may be lost to British Columbia and Canada because of financial attractions and facilities available beyond our borders. It was his hope that this fellowship may be used to attract such abilities to British Columbia. Prominent consideration is to be given to Masters and Ph.D. candidates in Medicine, Languages, Engineering and Forestry, but not necessarily in that order.

☐ **#6503 - Kathleen Simpson Memorial Graduate Scholarship**



A \$2,500 scholarship has been endowed by the Estate of Kathleen Simpson for a graduate student studying social applied science.

Equivariant Reinforcement Learning Policies with Symmetry Breaking

Dominic Klukas — M.A.Sc., EECE, UBC — Supervisor: Prof. Cyrus Neary

Keywords: AI, Robotics, Reinforcement Learning, Geometric Deep Learning, Equivariance

Background and Motivation. Modern deep reinforcement learning (RL) provides promising solutions to sequential decision making tasks such as robotics, optimization, and controls problems; however they suffer from high sample complexity and lack of generalization. In deep learning, *Group Equivariant Neural Networks* leverage symmetries present in a problem space to improve sample efficiency and generalization by enforcing them as inductive biases [1] [2]. For example, symmetries arise in physics (e.g. conservation laws) [3] as well as robotics and RL tasks (optimal policies are often invariant to flips, translations, and rotations, for example Figure 1) [6] [7]. However, while these symmetries are often present in RL tasks, in practice they are almost always approximate or even broken explicitly, thus rendering equivariant learning ineffective [5] [9].

Methods and Objectives. My Masters research project will address this key challenge by developing equivariant RL algorithms that are sample efficient, generalize well, and are robust to symmetry breaking. To do so, I will combine techniques from mathematical analysis, geometric learning, RL, and neural network design to pursue the following research objectives. Together, these three objectives will help bring the data efficiency of geometric deep learning to real world problems by relaxing their overly-strict assumptions of hard symmetries.

(O1) *Detecting local explicit symmetry breaking.* To effectively leverage symmetries in learning algorithms, we first must characterize where those symmetries apply in practice. My first objective will be to develop an algorithm using tools from hypothesis testing which can detect where within a state space those symmetries are broken in a robotics trajectory dataset [4].

(O2) *Developing locally equivariant NN architectures.* To capitalize on the knowledge of the overarching symmetry and where and how it is broken, the RL policies must be designed to obey them as inductive biases. Thus, my next objective will be to develop NN architectures that appropriately relax the equivariance constraints in equivariant neural networks so that the symmetries hold and can be broken in appropriate regions of the state space [8] [9].

(O3) *Equivariant RL for robotics.* Current work on equivariant RL primarily focuses on simple environments where task symmetries hold exactly, leading to promising but limited demonstrations. My last objective will be to investigate cases when locally equivariant models can outperform baselines in real robotic settings, where symmetries are only approximate due to robot kinematics, obstacles, and physical interactions. I will evaluate my policies in simulation using MuJoCo-based benchmarks against non-equivariant baselines, and on real robots in our lab.

Work Plan (Award Period). Throughout the course of my masters, I will work on (O1) in months 1-5, and (O2) in months 6-12, and I will seek to work on (O3) concurrently, but finish testing in months 13-18.

Significance and Expected Contributions. Unlike the internet scale-data used to train large-scale language and vision transformer models, the trajectory data used to train robotics is scarce. Furthermore, robot learning problems are complex, with slight modifications to tasks potentially resulting in catastrophic failures. The techniques developed through this masters will address both the sample efficiency and brittleness problems that plague modern RL and AI methods, enabling their application in robotics.

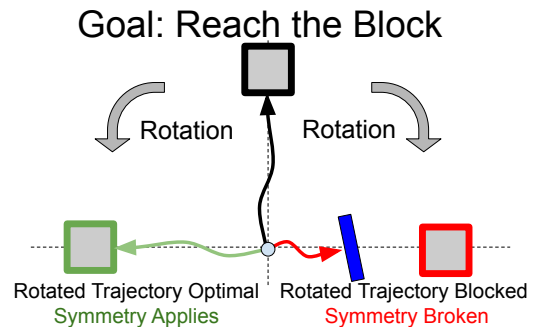


Figure 1: Equivariant policy and symmetry breaking. The task (reach the block) is equivariant under a counterclockwise rotation, but symmetry is broken under a clockwise rotation because of an obstacle.

Bibliography

- [1] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges. 4 2021.
- [2] Taco Cohen and Max Welling. Group equivariant convolutional networks. In Maria Florina Balcan and Kilian Q. Weinberger, editors, *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2990–2999, New York, New York, USA, 20–22 Jun 2016. PMLR.
- [3] Pierre-Yves Lagrave and Eliot Tron. Equivariant neural networks and differential invariants theory for solving partial differential equations. *Physical Sciences Forum*, 5(1), 2022.
- [4] E.L. Lehmann and J.P. Romano. *Testing Statistical Hypotheses*. Springer Texts in Statistics. Springer New York, 2006.
- [5] Jung Yeon Park, Sujay Bhatt, Sihan Zeng, Lawson L. S. Wong, Alec Koppel, Sumitra Ganesh, and Robin Walters. Approximate equivariance in reinforcement learning. In *International Conference on Artificial Intelligence and Statistics*, 2024.
- [6] Joohwan Seo, Soochul Yoo, Junwoo Chang, Hyunseok An, Hyunwoo Ryu, Soomi Lee, Arvind Kruthiventy, Jongeun Choi, and Roberto Horowitz. Se(3)-equivariant robot learning and control: A tutorial survey. *International Journal of Control, Automation and Systems*, 23(5):1271–1306, 2025.
- [7] Dian Wang, Mingxi Jia, Xupeng Zhu, Robin Walters, and Robert Platt. On-robot learning with equivariant models. In Karen Liu, Dana Kulic, and Jeff Ichnowski, editors, *Proceedings of The 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pages 1345–1354. PMLR, 14–18 Dec 2023.
- [8] Rui Wang, Elyssa Hofgard, Han Gao, Robin Walters, and Tess E. Smidt. Discovering symmetry breaking in physical systems with relaxed group convolution. In *Proceedings of the 41st International Conference on Machine Learning*, ICML’24. JMLR.org, 2024.
- [9] Rui Wang, Robin Walters, and Rose Yu. Approximately equivariant networks for imperfectly symmetric dynamics. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23078–23091. PMLR, 17–23 Jul 2022.



TRANSCRIPT OF ACADEMIC RECORD

Surname:
Klukas

Given Name:
Dominic Herbert

Student Number:
64348378

Date of Birth:
February 9

Date Printed:
November 26, 2025

Issued To:
Dominic Klukas
klukasdominic@gmail.com

REGISTRAR

Note: Transcript valid only if bearing the registrar's signature. This transcript is printed on secure paper on a blue background.

UBC Credentials

Credential: Bachelor of Applied Science
Program(s) of Study: In Engineering Physics
Minor in Honours Math
Honours/Awards: with Distinction
Granted: May 14, 2025

Transfer Credits

None to date

2019-20 Winter Session

Bachelor of Applied Science (UBC Vancouver) - Year 1

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	APSC_V 100	(3.0)	Introduction to Engineering I	85	A	3.0	216	77
1	CHEM_V 154	(3.0)	CHEM FOR ENGR	90	A+	3.0	312	72
1	ECON_V 101	(3.0)	Principles of Microeconomics	92	A+	3.0	134	71
1	ENGL_V 112	(3.0)	Strategies for University Writing	81	A-	3.0	30	77
1	MATH_V 120	(4.0)	Honours Differential Calculus	73	B	4.0	31	71
1	PHYS_V 157	(3.0)	Introductory Physics for Engineers I	89	A	3.0	283	78
2	APSC_V 101	(3.0)	Introduction to Engineering II	80	A-	3.0	162	77
2	APSC_V 160	(3.0)	Introduction to Computation in Engineering Design	96	A+	3.0	217	83
2	MATH_V 101	(3.0)	Integral Calculus with Applications to Physical Sciences and Engineering	82	A-	3.0	65	71
2	MATH_V 152	(3.0)	Linear Systems	94	A+	3.0	188	78
2	PHYS_V 158	(3.0)	Introductory Physics for Engineers II	90	A+	3.0	331	69
2	PHYS_V 159	(1.0)	Introductory Physics Laboratory for Engineers	85	A	1.0	53	83
2	PHYS_V 170	(3.0)	Mechanics I	100	A+	3.0	321	84

Dean's List

As of 16 March 2020, the COVID-19 pandemic disrupted regular academic activities. Modes of instruction and assessment were shifted to on-line activities mid-term, including changes to exam practices and weighting in some cases. Deadlines to withdraw or change to Credit/D/Fail or Pass/Fail grading were extended by some programs.

2020-21 Winter Session

Bachelor of Applied Science (UBC Vancouver) - Year 2

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	CPEN_V 221	(4.0)	Principles of Software Construction	88	A	4.0	260	71
1	ELEC_V 204	(4.0)	Linear Circuits	97	A+	4.0	125	96
1	ENPH_V 259	(3.0)	Experimental Techniques	82	A-	3.0	62	88
1	MATH_V 217	(4.0)	Multivariable and Vector Calculus	91	A+	4.0	101	75
1	MATH_V 255	(3.0)	Ordinary Differential Equations	90	A+	3.0	55	90
2	JAPN_V 201	(3.0)	BEGIN JAPN II	93	A+	3.0	21	79
2	MATH_V 220	(3.0)	MATH PROOF	83	A-	3.0	70	64
2	MATH_V 257	(3.0)	Partial Differential Equations	90	A+	3.0	5	72
2	MECH_V 260	(3.0)	Introduction to Mechanics of Materials	100	A+	3.0	106	77
2	PHYS_V 301	(3.0)	Electricity and Magnetism	100	A+	3.0	19	77

Dean's List



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2020-21 Winter Session continued...

UBC Academic Awards

Trek Excellence Scholarship for Continuing Students

2021 Summer Session

Bachelor of Applied Science (UBC Vancouver) - Year 2

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
	ENPH_V 253	(5.0)	Introduction to Instrument Design	99	A+	5.0	58	88
	ENPH_V 270	(2.0)	Mechanics II	87	A	2.0	61	86
	PHYS_V 203	(4.0)	Thermal Physics I	97	A+	4.0	62	78
	PHYS_V 250	(3.0)	Introduction to Modern Physics	83	A-	3.0	64	82

2021-22 Winter Session

Bachelor of Applied Science (UBC Vancouver) - Year 3

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	ELEC_V 221	(4.0)	Signals and Systems	100	A+	4.0	122	78
1	ENPH_V 353	(3.0)	Engineering Physics Project I	93	A+	3.0	44	86
1	MATH_V 307	(3.0)	Applied Linear Algebra	93	A+	3.0	115	77
1	MATH_V 320	(3.0)	Real Variables I	94	A+	3.0	38	60
1	MECH_V 360	(3.0)	Mechanics of Materials	86	A	3.0	131	68
2	APSC_V 278	(3.0)	Engineering Materials	88	A	3.0	237	72
2	APSC_V 279	(1.0)	Engineering Materials Laboratory	89	A	1.0	14	90
2	CPEN_V 312	(3.0)	Digital Systems and Microcomputers	89	A	3.0	163	80
2	MATH_V 305	(3.0)	Applied Complex Analysis	99	A+	3.0	71	77
2	MECH_V 280	(3.0)	Introduction to Fluid Mechanics	94	A+	3.0	54	77
2	PHYS_V 304	(3.0)	Introduction to Quantum Mechanics	92	A+	3.0	92	76
2	PHYS_V 350	(3.0)	Applications of Classical Mechanics	94	A+	3.0	64	85

Dean's List

UBC Academic Awards

Donald J. Evans Scholarship in Engineering
Trek Excellence Scholarship for Continuing Students

2022-23 Winter Session

Bachelor of Applied Science (UBC Vancouver) - Year 4

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	ELEC_V 341	(4.0)	Systems and Control	94	A+	4.0	40	66
1	MATH_V 322	(3.0)	Introduction to Group Theory	77	B+	3.0	36	61
1	MATH_V 400	(3.0)	Applied Partial Differential Equations	90	A+	3.0	48	79
1	MATH_V 437	(3.0)	Number Theory	94	A+	3.0	18	81
1	MECH_V 325	(4.0)	Machine Design	81	A-	4.0	95	79
1-2	ENPH_V 459	(5.0)	ENG PHYS PRJ2	89	A	5.0	60	86
2	APSC_V 450	(2.0)	Professional Engineering Practice	80	A-	2.0	305	83
2	CPSC_V 221	(4.0)	Basic Algorithms and Data Structures	90	A+	4.0	144	82
2	MATH_V 318	(3.0)	Probability with Physical Applications	95	A+	3.0	104	78
2	MATH_V 443	(3.0)	Graph Theory	95	A+	3.0	30	91
2	PHYS_V 401	(3.0)	Electromagnetic Theory	81	A-	3.0	82	84

Dean's List

UBC Academic Awards

Donald J. Evans Scholarship in Engineering
Trek Excellence Scholarship for Continuing Students

2023 Summer Session

Bachelor of Applied Science (UBC Vancouver) - Year 4



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2023 Summer Session continued...

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
	APSC_V 110	(3.0)	Co-operative Education Work Term I		P	3.0		

2023-24 Winter Session

Bachelor of Applied Science (UBC Vancouver) - Year 5

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	APSC_V 210	(3.0)	Co-operative Education Work Term		P	3.0		
2	MATH_V 321	(3.0)	Real Variables II	90	A+	3.0	37	67
2	MATH_V 323	(3.0)	Introduction to Rings and Modules	77	B+	3.0	22	58
2	MATH_V 401	(3.0)	Green's Functions and Variational Methods	87	A	3.0	16	82
2	PHYS_V 403	(3.0)	Statistical Mechanics	94	A+	3.0	113	77

2024 Summer Session

Bachelor of Applied Science (UBC Vancouver) - Year 5

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
	APSC_V 310	(3.0)	Co-operative Education Work Term		P	3.0		

2024-25 Winter Session

Bachelor of Applied Science (UBC Vancouver) - Year 5

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	ENPH_V 352	(2.0)	Laboratory Techniques in Physics	90	A+	2.0	20	90
1	MATH_V 418	(3.0)	Probability		W			
1	MATH_V 420	(3.0)	Real Analysis I	91	A+	3.0	22	85
1	MATH_V 422	(3.0)	Fields and Galois Theory	93	A+	3.0	8	95
1	MATH_V 449D	(3.0)	Honours Reading	100	A+	3.0	13	99
1-2	ENPH_V 479	(6.0)	Engineering Physics Project III	83	A-	6.0	69	84
2	CHIN_V 233	(3.0)	Basic Chinese II: Part 2 (Non-Heritage)	86	A	3.0	24	73
2	CIVL_V 250	(3.0)	Engineering and Sustainable Development	82	A-	3.0	220	88
2	ELEC_V 302	(3.0)	Electronic Circuits for Electromechanical Design	83	A-	3.0	74	78
2	ELEC_V 481	(3.0)	Economic Analysis of Engineering Projects	81	A-	3.0	67	70
2	PHYS_V 408	(4.0)	Optics	92	A+	4.0	96	77

Cumulative Totals

Credits Earned: 211.0 Average: 89.6%

Dean's List

UBC Academic Awards

John Collison Memorial Scholarship in Mathematics

2025-26 Winter Session

Master of Applied Science (UBC Vancouver)

Term	Course	Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	EECE_V 571N	(3.0)	Sequential Decision Making		CIP			
1	ELEC_V 571	(3.0)	Control Systems		CIP			
1	MATH_V 544	(3.0)	Probability I		CIP			
1-2	EECE_V 595C	(2.0)	Electrical and Computer Engineering Seminar		CIP			
1-2	EECE_V 599	(12.0)	Thesis		CIP			
2	EECE_V 571F	(3.0)	Advanced Topics in Deep Learning					
2	EECE_V 589	(3.0)	Introduction to Robotics					
2	MATH_V 510	(3.0)	Functional Analysis					
2	MATH_V 545	(3.0)	Probability II					

Cumulative Totals

Credits Earned: 0.0 Average: 0.0%



TRANSCRIPT OF ACADEMIC RECORD

Surname:
Klukas

Given Name:
Dominic Herbert

Student Number:
64348378

Date of Birth:
February 9

Date Printed:
November 26, 2025

***** End of Record *****





The University of British Columbia was established by an Act of the Provincial Legislature in 1908 and opened in 1915. The University is a member of the Universities Canada. UBC Okanagan was established as a campus of The University of British Columbia in April 2005. UBC Okanagan assumed responsibility for several Okanagan University College degree programs and continuing students in those programs on July 1, 2005. Okanagan University College ceased its educational operations as of August 31, 2005.

EXPLANATION OF STUDENT ACADEMIC RECORD

Academic terms

The University operates on a year system. The academic year is divided into two sessions. Winter Session is divided into two terms, Term 1 from September to December and Term 2 from January to April. Summer Session starts in May and ends in August. Winter session courses may span a single term (Term 1 or 2) or a full session (Term 1 and 2).

Distance Education and Technology (DET) courses begin on one of two starting dates during Winter Session, designated in the Term column as A,B,C or D. During the Summer Session, DET courses have only one starting date and are designated with an A or B in the Term column. DET courses ceased to be offered as of May 2022.

Academic Awards

All academic awards (Fellowships, Scholarships, Prizes and Medals) administered by the University and issued subsequent to April 1975, are noted on the transcript of academic record.

Course Codes

Courses starting July 1, 2005 onwards, a suffix has been added that indicates the campus that is offering the course. _V for courses offered by the Vancouver campus and _O for courses offered by the Okanagan campus.

Course Numbering

Courses numbered 100 to 199 are primarily for the first year students. Similarly, courses numbered 200 to 299 are primarily for second year students, courses numbered 300 to 399 are for third year students, and courses numbered 400 to 499 are for fourth year students. Courses numbered 500 to 699 are considered graduate-level. Courses numbered 700 and above are residency courses.

Class Size and Class Average Grades

Effective February 1999, class size and class average grades are shown for most courses. In courses with more than one section, the class size and average are reported for the student's section.

Grading and Course Weight

In May of 1991, The University of British Columbia adopted a percentage grading and credit weighting system. Course weight is expressed in "credits". In general 1 credit represents 1 hour of instruction or 2 to 3 hours of laboratory work per week throughout one term of the Winter Session. A 1-unit course became a 2-credit course. Courses are normally graded on a percentage basis with a corresponding letter grade derived as shown.

Degree Standings

Standings are expressed as with Honours, with Distinction or with High Distinction; not used by all faculties. Calculation of the standing is dependent on the Faculty. Refer to the UBC Academic Calendar for more information. This is not applicable to most graduate programs.

Cumulative Average and Cumulative Credits

Effective September 2024, cumulative average and cumulative credits are shown after each Academic Period. Students with multiple degrees, the cumulative average and credits are shown independently for each degree.

Transfer Credit

Transfer credit totals include all transfer credit awarded. Most undergraduate faculties allow a maximum of 50% of degree requirements to be fulfilled by transfer credit awarded.

Undergraduate Grading Scale

Percent	Letter Grade	Percent	Letter Grade
90–100	A+	64–67*	C+
85–89	A	60–63*	C
80–84	A-	55–59*	C-
76–79	B+	50–54*	D
72–75	B	0–49	F* (Fail)
68–71	B-		

Graduate Grading Scale

Masters Grading Scale		Doctoral Grading Scale	
Percent	Letter Grade	Percent	Letter Grade
90–100	A+	90–100	A+
85–89	A	85–89	A
80–84	A-	80–84	A-
76–79	B+	76–79	B+
72–75	B	72–75	B
68–71	B-	68–71	B-
64–67	C+		
60–63	C	0–67	F* (Fail)
0–59	F* (Fail)		

*Some programs may require higher passing grades; failing grades are assigned a letter grade of "F".

The following standings also apply:

- AEG
 - aegrotat; allows a student credit for a course even though the course requirements have not been completed due to medical, emotional or other difficulties; a letter grade is assigned and converted to the minimum percentage for that category for the calculation of averages
 - audit; granted to students who have been approved as having auditing status
- CH
 - challenge credit; granted to a student that gives course credit when the student meets the minimum required percentage grade on a challenge exam
- CIP
 - course in progress or without a grade submitted before the grade submission deadline

CR/D/F

- Credit/D/Fail excluded from calculation in all averages; granted where a grade is assigned by an instructor but not included on the student's official transcript; grade is converted to CR, D or F standing. CR– credit, satisfactory completion of the requirements of the subject; D– marginal pass (50%–54%), and not available for those programs or courses that define a passing grade as higher than 50%; F– fail

EX

- exempt; granted to students who have registered for a required course from which they are exempt

EXG

- externally graded; nominal standing used administratively to maintain registration in a UBC program of study while the student is taking and receiving a grade for a course at another institution (introduced September 2024)

J

- adjudicated pass; granted where credit is granted and the course need not be repeated although it may not normally qualify as a prerequisite for further work; used in the calculation of averages

GNS

- grade not submitted; course for which the deadline for grade submission has passed and a grade was not submitted (introduced September 2024)

P or F

- no grade assigned, excluded from calculation in all averages; P–pass, requirements of subject completed satisfactorily; credit granted where applicable; F– fail

- honours; exceeds course requirements, credit granted where applicable. No grade assigned, excluded from calculation in all averages

H

- prior learning assessment; credit assigned based on prior learning in subject area

- supplemental; standing that indicates that the student has passed on the basis of a supplemental examination. Credit has been earned

PLA

- standing deferred; excluded from calculation in all averages

- thesis in progress; graduating essay not submitted; course continuing

- official withdrawal

SD

T

W

For courses taken prior to May 1991, refer to the following website for descriptions of the transcript elements.

<https://vancouver.calendar.ubc.ca/campus-wide-policies-and-regulations/transcript-academic-record>

THIS DOCUMENT CONTAINS SECURITY FEATURES. HOLD AT AN ANGLE TO VERIFY A WATERMARK BASKET WEAVE DESIGN



Canadian Institutes of
Health Research

Natural Sciences and Engineering
Research Council of Canada

Social Sciences and Humanities
Research Council of Canada

Instituts de recherche en
santé du Canada

Conseil de recherches en sciences
naturelles et en génie du Canada

Conseil de recherches en
sciences humaines du Canada

Canada



Protected when completed

Date Submitted: 2025-11-28 14:55:06

Confirmation Number: 2034913

Template: CGRS-Master's

Mr. Dominic Klukas

Correspondence language: English

Date of Birth: 2/09

Canadian Residency Status: Canadian Citizen

Country of Citizenship: Austria, Canada

Contact Information

The primary information is denoted by (*)

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Delta British Columbia V4K1J5
Canada

Telephone

Mobile (*) 672-999-3141

Email

Personal (*) klukasdominic@gmail.com

Mr. Dominic Klukas

Language Skills

Language	Read	Write	Speak	Understand
Chinese	Yes	Yes	Yes	Yes
English	Yes	Yes	Yes	Yes
German	Yes	Yes	Yes	Yes
Japanese	Yes	Yes	Yes	Yes

User Profile

Research Disciplines: Computer Engineering and Software Engineering

Areas of Research: Algorithms

Fields of Application: Industrial Manufacturing and Production

Research Specialization Keywords: Geometric Learning, Reinforcement Learning, Robotics

Degrees

2025/9	Master's Thesis, Electrical and Computer Engineering, Optimization, University of British Columbia Degree Status: In Progress
2019/9 - 2025/5	Bachelor's, Applied Science, Engineering Physics, Minor in Honours Mathematics, 211 Credits, AVG 90, University of British Columbia Degree Status: Completed

Recognitions

2024/9	John Collinson Memorial Scholarship in Mathematics - 6,050 (Canadian dollar) University of British Columbia Prize / Award
2023/9	Eric A. Roenitz Memorial Award in Engineering Physics - 500 (Canadian dollar) University of British Columbia Prize / Award
2022/12	Donald J. Evans Scholarship in Engineering - 500 (Canadian dollar) University of British Columbia Prize / Award
2022/9	Trek Excellence Scholarship - 1,500 (Canadian dollar) University of British Columbia Prize / Award

2021/12	Donald J. Evans Scholarship in Engineering - 500 (Canadian dollar) University of British Columbia Prize / Award
2021/9	Trek Excellence Scholarship - 1,500 (Canadian dollar) University of British Columbia Prize / Award
2020/9	Trek Excellence Scholarship - 1,500 (Canadian dollar) University of British Columbia Prize / Award

Employment

2024/5 - 2024/8	Product Engineering Co-op. Characterization of high performance data center crossbar switches. Built python testing frameworks. Microchip Technology
2023/5 - 2023/12	Test Engineering Co-op. Mechanical Design, Electronics Assembly, Testing, and Documentation. ARTMS Inc.
2022/5 - 2022/8	Research Assistant. Sauder school of business. Wrote numerical analysis for paper: "Performance of the Offer-Everything Policy", Journal of Operations Research University of British Columbia
2020/6 - 2020/8	Management Internship. Wrote batch and VBA scripts to automate office work. Created spreadsheet applications to solve operations problems, using VBA and SQL. Siemens Mobility Austria GmbH

Research Funding History

Awarded [n=1]

2022/5 - 2022/8 Principal Applicant	NSERC Undergraduate Student Research Award
	Funding Sources:
	2022/5 - 2023/8 Natural Sciences and Engineering Research Council of Canada (NSERC)
	Total Funding - 6,000 (Canadian dollar)
	Funding Competitive?: Yes